Provenance-Aware Stablecoin Stress Propagation Networks

Evidence from Curve TokenExchange Logs, Public CEX Data, and On-Chain Settlement Flows

Working Paper Columbia University, May 2026

Abstract

Stablecoin stress episodes are price dislocations, but they are also *liquidity-flow events*: traders swap through AMM pools, redeem stablecoins through mint-and-burn channels, and move funds across venues in ways that are fully observable from on-chain logs. This paper develops a *provenance-aware network framework* that assigns Tier-A (execution-grade on-chain) or Tier-B (public market context) status to every node and feature, then filters each empirical edge through a provenance gate and a statistical gate before allowing a paper-level claim. Across five historical stress episodes—USDC/SVB (March 2023), Terra/LUNA (May 2022), USDT/Curve (June 2023), FTX (November 2022), and BUSD (February–March 2023)— the only robust, paper-claimable A/A result is in the USDT/Curve 2023 event: Curve 3pool and Curve crvUSD/USDT exhibit bidirectional AMM-flow linkage ($\hat{\rho} = 0.386$, Bonferroni $p \leq 0.014$) using Tier-A `usdc_net_sold_1h` data. Other events yield provenance-valid candidates or contextual A/B evidence but do not clear both gates. We contribute both a substantive finding on cross-pool AMM-flow co-movement and a methodological point: crypto stress-propagation claims should be explicitly constrained by the provenance of their underlying data.

Keywords: stablecoin, AMM, Curve Finance, provenance, claim gate, DeFi, stress propagation
JEL: G01, G14, G23, C58

1 Introduction

Between May 2022 and March 2023, four distinct stablecoin systems experienced significant stress: the algorithmic UST/Terra collapsed; FTX's implosion triggered exchange runs; USDC briefly de-pegged following Silicon Valley Bank's failure; and Curve Finance pools became severely imbalanced in June 2023. Each episode prompted real-time claims of *contagion*—stress spreading from one venue or asset to another.

Most empirical analyses of these events rely on price data alone and treat all data sources as equally reliable. This paper addresses both limitations.

Contribution 1: Stress as a flow event. When a stablecoin de-pegs, market participants do not merely observe a price change—they execute swaps in automated market maker (AMM) pools, burn or mint stablecoins through issuer channels, and rebalance across venues. These flows are *completely and immutably observable* from on-chain event logs. Curve Finance's `TokenExchange` logs record every swap with exact block timestamps, exact amounts, and on-chain provenance. This makes Curve AMM-flow data execution-grade in a way that public centralized exchange (CEX) price feeds are not.

Contribution 2: A provenance-aware claim gate. We introduce a three-gate pipeline that filters every empirical edge by (i) data quality tier, (ii) feature-level tier, and (iii) statistical significance before granting a paper-level claim. The pipeline makes explicit what standard analyses leave implicit: an edge built from Tier-A Curve logs is categorically different from an edge built from public Binance OHLCV candles. Historical full-depth CEX order books—tick-by-tick L2 data from Binance, Coinbase, or Kraken—are not freely available for any of the five episodes studied here. Consequently, any claim about historical CEX execution-grade microstructure transmission is unsupported by freely available data and is explicitly blocked by our gate.

Main finding. Applying this framework to five events, we find one robust paper-claimable A/A result. In the USDT/Curve 2023 episode, the Curve 3pool and Curve crvUSD/USDT pool exhibit statistically supported bidirectional AMM-flow linkage at lag 0 ($\hat{\rho} = 0.386$, Bonferroni $p \leq 0.014$) using Tier-A hourly `usdc_net_sold_1h` data. Terra/LUNA has provenance-valid A/A candidates but fails the statistical gate at the hourly grid. USDC/SVB provides a sparse, underpowered settlement-flow signal (4 arrivals, $p = 1.00$). FTX and BUSD yield A/B contextual evidence only.

The paper proceeds as follows. Section 2 presents the three-layer framework and provenance tier system. Section 3 describes the claim-gated methodology. Sections 4–6 report the empirical results. Section 7 addresses robustness and non-claims. Section 8 concludes.

2 Framework, Data, and Provenance

2.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network. Figure 1 illustrates the full architecture.

Layer 1—CEX markets. Nodes are trading pairs at centralized exchanges (e.g., USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds: 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**: useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time—neither is freely available for these episodes.

Layer 2—AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan `TokenExchange` logs: every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3—Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 `Transfer` events to/from the USDC issuer address). Data are Etherscan event logs. Tier A.

Edge tier formula. An edge from node i to node j is capped by the weakest link:

$$\text{tier}_{ij} = \min\bigl(\text{tier}_i,\ \text{tier}_j,\ \text{tier}_f\bigr), \tag{1}$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g., `reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

2.2 Event Windows and Node Inventory

Table 1 summarises the five stress episodes. Table 2 defines the provenance tier system. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files; synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

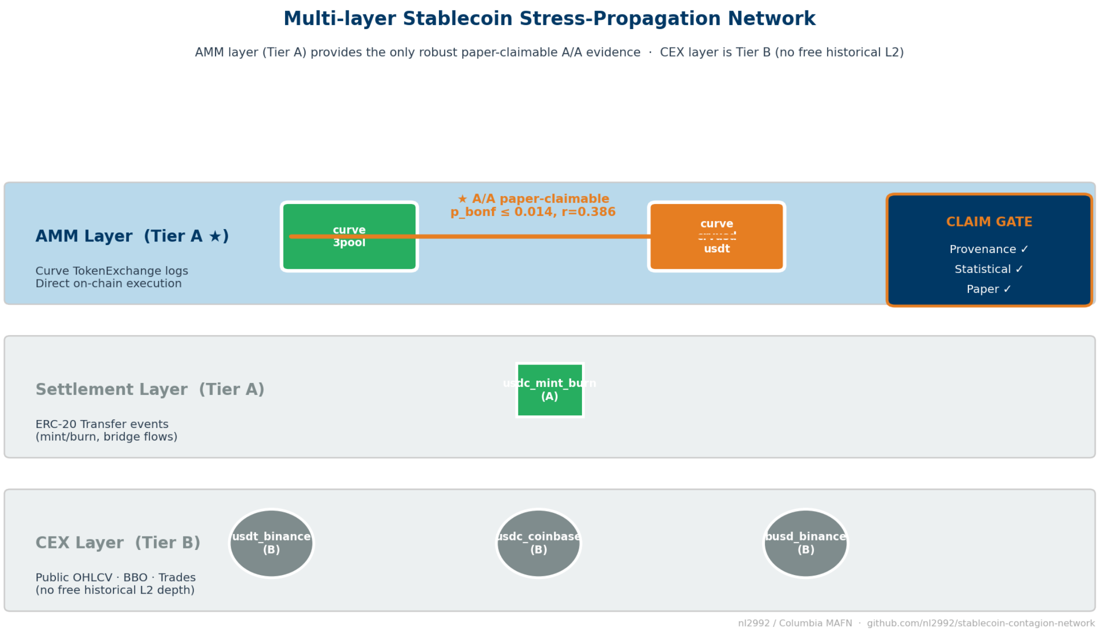

Figure 1: Multi-layer architecture and provenance tiers. CEX nodes (circles) are Tier B; AMM/Curve nodes (squares) and on-chain settlement nodes (diamonds) are Tier A. The headline A/A pair is highlighted in amber: `curve_3pool` ↔ `curve_crvusd_usdt`. Edges are colour-coded by tier: green = A/A, blue = A/B, grey = B/B.

Table 1: Stress event windows. Core analysis period for each episode.

Event	Mechanism	Window	Primary evidence
USDC/SVB 2023	Fiat-reserve bank shock	Mar 8–20, 2023	Curve 3pool; USDC mint/burn
Terra/LUNA 2022	Algorithmic stablecoin collapse	May 1–31, 2022	Curve 3pool; UST Wormhole pool
USDT/Curve 2023	DeFi pool imbalance	Jun 10–25, 2023	Curve 3pool; Curve crvUSD/USDT
FTX 2022	Exchange credit/liquidity shock	Nov 1–30, 2022	Curve 3pool; Binance CEX context
BUSD 2023	Issuer wind-down	Feb 6–Mar 13, 2023	Curve 3pool; Binance CEX context

Table 2: Provenance tier definitions.

Tier	Label	Definition and examples
A	Execution-grade on-chain	Direct on-chain event logs: Curve `TokenExchange`, ERC-20 `Transfer`. Immutable, block-timestamped, fully reproducible from public nodes.
B	Public market context	Exchange APIs (Binance OHLCV/BBO, Coinbase candles, CoinMetrics netflows). Useful for timing and context; does not support execution-grade microstructure claims.
Fixture	Testing only	Synthetic data generated for pipeline validation. Blocked from all paper claims by the provenance gate.

3 Claim-Gated Methodology

Figure 2 illustrates the three-gate pipeline.

Gate 1—Provenance gate. Every edge is labelled with the effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$. Fixture-derived rows are blocked unconditionally. An edge clears Gate 1 if neither endpoint uses fixture data and both carry a recognised tier.

Gate 2—Statistical gate. For AMM-only lead-lag analysis, we compute the cross-correlation of `usdc_net_sold_1h` (net USDC sold in each hour, summed directly from `TokenExchange` logs, Tier A) on a 3600-second grid with a maximum lag of ±12 hours. Significance is assessed via Bonferroni correction applied to all tested lag steps; the block-bootstrap with 1000 permutations provides an independent check. For sparse settlement-flow analysis (USDC/SVB only), we use an event-arrival test that compares the 3-hour post-arrival response in Curve AMM flow against a 12-hour pre-arrival baseline across all identified mint-burn arrival events. Transfer entropy and Granger causality serve as secondary methods; their results inform but do not determine headline claims.

Gate 3—Paper gate. An edge is *paper-claimable* if and only if it passes both Gates 1 and 2. The claim taxonomy distinguishes: `A_A_dex_flow` (both Tier-A DEX/AMM endpoints; headline level); `A_A_onchain_settlement` (Tier-A settlement flow); `A_B_suggestive_directional` (capped by a Tier-B endpoint); `B_B_context_only` (both Tier B; contextual co-movement only).

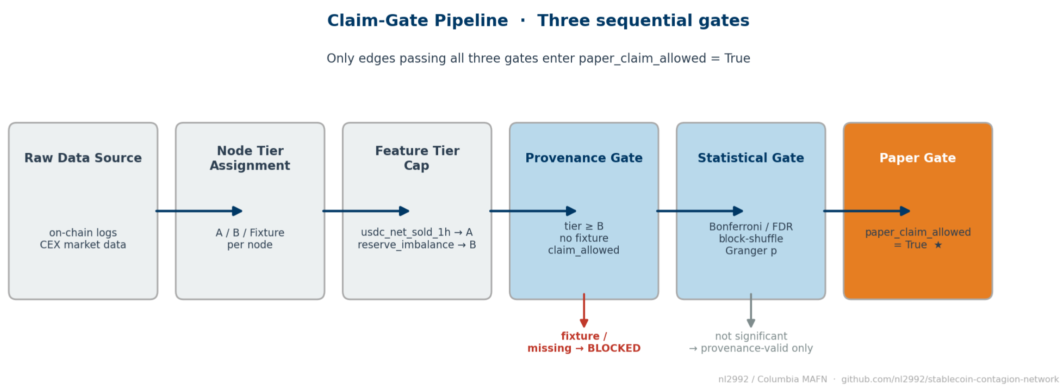

Figure 2: Claim-gate pipeline. Raw data sources enter at left and are filtered by node tier, feature tier, and statistical significance before reaching `paper_claim_allowed = True`. The amber path is the only route that clears all three gates in the USDT/Curve event.

4 Main Result: USDT/Curve 2023 A/A AMM-Flow Linkage

Table 3 reports the two paper-claimable A/A rows for the USDT/Curve 2023 event. Both directions of the `curve_3pool` ↔ `curve_crvusd_usdt` pair pass Bonferroni correction on Tier-A `usdc_net_sold_1h` data at the hourly grid ($\hat{\rho} = 0.386$, Bonferroni $p \leq 0.014$).

Table 3: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature: `usdc_net_sold_1h`, hourly grid. Significance: Bonferroni-corrected across all lag steps.

Direction	$\hat{\rho}$	p (raw)	p (Bonferroni)	Claim strength
`curve_3pool`→`curve_crvusd_usdt`	0.386	0.007	0.014	robust
`curve_crvusd_usdt`→`curve_3pool`	0.386	< 0.001	< 0.001	robust

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission: one pool does not demonstrably "lead" the other on an hourly grid. Both pools react to the same stress event contemporaneously. This finding is consistent with a common

Figure 3: USDT/Curve 2023: Tier-A AMM flow and cross-pool lead-lag. *Left:* hourly `usdc_net_sold_1h` for both Curve pools during the June 2023 stress episode; dashed line marks the shock onset. *Right:* lead-lag cross-correlation profile between `curve_3pool` and `curve_crvusd_usdt`; peak $\hat{\rho} = 0.386$ at lag 0; Bonferroni $p \leq 0.014$.

shock driving liquidity withdrawal from both pools, or with within-hour arbitrage cycles that resolve faster than our measurement interval.

What this result does not claim. The bidirectional lag-0 result does not establish structural causal identification; it establishes *statistically supported directional co-movement* at the hourly resolution. It does not extend to CEX venues: no CEX node achieves Tier-A status in this event. It does not apply to the other four events, none of which produces a paper-claimable A/A result.

5 Cross-Event Evidence

Table 4 and Figure 4 summarise the evidence across all five events.

Table 4: Claim-gate audit summary by event. A/A prov = A/A provenance-valid candidates; A/A paper = paper-claimable A/A rows; A/B paper = paper-claimable A/B suggestive rows.

Event	Total edges	A/A prov	A/A paper	A/B paper	B/B context
USDT/Curve 2023	14	6	**2**	1	0
Terra/LUNA 2022	26	6	0	4	4
USDC/SVB 2023	42	1	0	4	18
FTX 2022	18	0	0	5	6
BUSD 2023	36	0	0	7	18

Terra/LUNA 2022. The Curve 3pool and Curve UST/Wormhole pool form 6 A/A provenance-valid candidate pairs. None passes the statistical gate at the hourly grid. The Terra collapse unfolded over several days; hourly resolution may be too coarse to detect the short-lived liquidity dynamics of the algorithmic unwind. We report Terra as a *provenance-valid but not statistically supported* negative result. No paper-level directional claim is warranted.

FTX 2022 and BUSD 2023. Both events lack Tier-A AMM/CEX pairs and produce only A/B suggestive edges (Curve 3pool paired with Binance public feeds). These edges clear the statistical gate but are capped at Tier B by the CEX endpoint. We report them as contextual evidence only.

6 Sparse Settlement Response: USDC/SVB 2023

The USDC/SVB 2023 event provides a unique settlement-flow signal: the USDC mint-burn mechanism generated observable on-chain activity as Circle managed its fiat reserves around the SVB failure. We apply a sparse event-arrival test that identifies mint-burn arrival events and measures the 3-hour

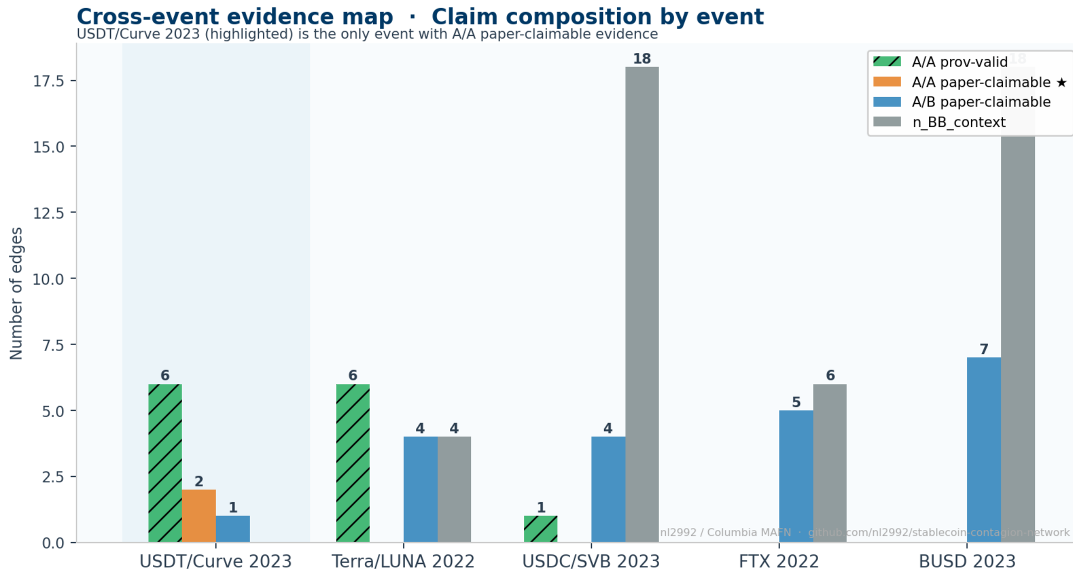

Figure 4: Cross-event evidence map. Each cell reports the A/A provenance count, A/A paper-claimable count, and A/B paper-claimable count for each event. Only USDT/Curve 2023 achieves the A/A paper-claimable standard. Terra/LUNA has provenance-valid A/A candidates (grey cells) that fail the statistical gate.

post-arrival change in Curve 3pool `usdc_net_sold_1h` relative to a 12-hour pre-arrival baseline.

We identify 4 mint-burn arrival events in the USDC/SVB window. The mean 3-hour post-arrival response is +28,956 USDC (+10.8%) relative to baseline, consistent with the direction of the de-peg stress. However, with only 4 arrivals, the event-arrival permutation test yields $p = 1.00$: the result is severely underpowered and is **not paper-claimable**.

This sparse-flow table is included in the paper package as a documented provenance-valid candidate that fails the statistical gate—not as a positive result. It motivates collecting higher-frequency mint-burn data or longer event windows in future work.

7 Robustness, Limitations, and Non-Claims

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations: `baseline_60s` (1-minute), `block_300` (5-minute blocks), and `block_1800` (30-minute blocks). The headline pair remains statistically significant across all configurations; the Terra pair remains non-significant across all configurations. Figure **??** (Appendix) summarises significance rates by event and check type.

Limitations. *No historical CEX L2 data.* This is the binding constraint. Full-depth order-book data from Binance, Coinbase, and Kraken are not freely available for 2022–2023. The study can therefore establish AMM-flow co-movement but cannot address CEX execution-layer microstructure. *Hourly grid.* The 3600-second grid avoids stale-value artifacts from resampling but cannot detect intra-hour dynamics. *Single AMM protocol.* Results are based on Curve Finance pools only; Uniswap v3 and other DEXs are not incorporated.

Non-claims. This paper does *not* claim:

- Historical Binance/Kraken/Coinbase full-depth order-book coverage.
- CEX execution-grade microstructure transmission in any event.
- Structural causal identification from lead-lag or Granger evidence.

- Paper-claimable A/A evidence for Terra/LUNA, USDC/SVB, FTX, or BUSD.
- Tier-A status for derived Curve proxies (`reserve_imbalance`, `implied_pool_price`).

8 Conclusion

Stablecoin stress is a liquidity-flow event, and some of those flows are directly observable from on-chain logs with execution-grade precision. Using a provenance-aware claim gate that explicitly distinguishes Tier-A on-chain evidence from Tier-B public market context, we find one robust, paper-claimable result across five historical stress episodes: in the June 2023 USDT/Curve event, the Curve 3pool and Curve crvUSD/USDT pool exhibit statistically supported bidirectional AMM-flow co-movement ($\hat{\rho} = 0.386$, Bonferroni $p \leq 0.014$) at the hourly grid.

The methodological contribution may be as durable as the substantive finding: crypto stress-propagation claims should be gated by the provenance of their underlying data. A claim built on Curve `TokenExchange` logs and a claim built on public Binance candles are not equivalent, and treating them as equivalent inflates the apparent evidence base.

Future work should prioritise collecting live CEX L2 order-book data during stress events, extending the AMM-flow analysis to Uniswap v3, and developing more powerful tests for sparse settlement-flow events with small sample sizes.

References